

BPL_STEM_AIR_Perfusion script with PyFMI

The key library PyFMI is installed.

After the installation a small application BPL_STEM_AIR_Pefusion is loaded and run. You can continue with this example if you like.

```
In [1]: !lsb_release -a # Actual VM Ubuntu version used by Google
```

```
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 22.04.4 LTS
Release:        22.04
Codename:       jammy
```

```
In [2]: %env PYTHONPATH=
```

```
env: PYTHONPATH=
```

```
In [3]: !python --version
```

```
Python 3.11.11
```

```
In [4]: !wget https://repo.anaconda.com/miniconda/Miniconda3-py311_24.11.1-0-Linux-x86_64.s
!chmod +x Miniconda3-py311_24.11.1-0-Linux-x86_64.sh
!bash ./Miniconda3-py311_24.11.1-0-Linux-x86_64.sh -b -f -p /usr/local
import sys
sys.path.append('/usr/local/lib/python3.11/site-packages/')
```

```
--2025-03-27 08:08:53-- https://repo.anaconda.com/miniconda/Miniconda3-py311_24.11.
1-0-Linux-x86_64.sh
Resolving repo.anaconda.com (repo.anaconda.com)... 104.16.32.241, 104.16.191.158, 26
06:4700::6810:20f1, ...
Connecting to repo.anaconda.com (repo.anaconda.com)|104.16.32.241|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 145900576 (139M) [application/octet-stream]
Saving to: 'Miniconda3-py311_24.11.1-0-Linux-x86_64.sh'
```

```
Miniconda3-py311_24 100%[=====>] 139.14M 106MB/s in 1.3s
```

```
2025-03-27 08:08:55 (106 MB/s) - 'Miniconda3-py311_24.11.1-0-Linux-x86_64.sh' saved
[145900576/145900576]
```

```
PREFIX=/usr/local
Unpacking payload ...
```

```
Installing base environment...
```

```
Preparing transaction: ...working... done
Executing transaction: ...working... done
installation finished.
```

```
In [5]: !conda update -n base -c defaults conda --yes
```

Channels:
- defaults
Platform: linux-64
Collecting package metadata (repodata.json): - 00\ 00| 00/ 00- 00\ 00| 00/ 00- 00\
00| 00/ 00done
Solving environment: \ 00| 00done

Package Plan

environment location: /usr/local

added / updated specs:
- conda

The following packages will be downloaded:

package	build	
-----	-----	
ca-certificates-2025.2.25	h06a4308_0	129 KB
certifi-2025.1.31	py311h06a4308_0	163 KB
openssl-3.0.16	h5eee18b_0	5.2 MB
-----	-----	
Total:		5.5 MB

The following packages will be UPDATED:

ca-certificates	2024.11.26-h06a4308_0 --> 2025.2.25-h06a4308_0
certifi	2024.8.30-py311h06a4308_0 --> 2025.1.31-py311h06a4308_0
openssl	3.0.15-h5eee18b_0 --> 3.0.16-h5eee18b_0

Downloading and Extracting Packages:

openssl-3.0.16	5.2 MB	: 0% 0/1 [00:00<?, ?it/s]
certifi-2025.1.31	163 KB	: 0% 0/1 [00:00<?, ?it/s]
ca-certificates-2025	129 KB	: 0% 0/1 [00:00<?, ?it/s]
openssl-3.0.16	5.2 MB	: 1% 0.005965187900324128/1 [00:00<00:17, 17.46 s/it]
ca-certificates-2025	129 KB	: 100% 1.0/1 [00:00<00:00, 6.19it/s]
certifi-2025.1.31	163 KB	: 39% 0.3936048816230436/1 [00:00<00:00, 3.53it/s]
ca-certificates-2025	129 KB	: 100% 1.0/1 [00:00<00:00, 6.19it/s]
certifi-2025.1.31	163 KB	: 100% 1.0/1 [00:00<00:00, 3.53it/s]

Preparing transaction: - 00done
Verifying transaction: | 00/ 00- 00done
Executing transaction: | 00done

```
In [6]: !conda --version  
        !python --version
```

```
conda 24.11.1  
Python 3.11.11
```

```
In [7]: !conda config --set channel_priority strict
```

```
In [8]: !conda install -c conda-forge pyfmi --yes # Install the key package
```

```
Channels:
- conda-forge
- defaults
Platform: linux-64
Collecting package metadata (repodata.json): - 22\ 22| 22/ 22- 22\ 22| 22/ 22- 22\
22| 22/ 22- 22\ 22| 22/ 22- 22\ 22| 22/ 22- 22\ 22| 22/ 22- 22\ 22| 22/ 22- 22\ 22|
22/ 22- 22\ 22| 22/ 22- 22\ 22| 22/ 22- 22\ 22done
Solving environment: / 22- 22\ 22done
```

Package Plan

environment location: /usr/local

added / updated specs:

- pyfmi

The following packages will be downloaded:

package	build		
-----	-----		
_x86_64-microarch-level-3	2_broadwell	8 KB	conda-forge
assimulo-3.6.0	py311h083bc19_0	1.1 MB	conda-forge
certifi-2025.1.31	pyhd8ed1ab_0	159 KB	conda-forge
conda-25.3.0	py311h38be061_0	1.1 MB	conda-forge
fmilib-2.4.1	hac33072_1	383 KB	conda-forge
gmp-6.3.0	hac33072_2	449 KB	conda-forge
libamd-3.3.3	haaf9dc3_7100102	49 KB	conda-forge
libblas-3.9.0	31_h59b9bed_openblas	16 KB	conda-forge
libbtf-2.3.2	h32481e8_7100102	27 KB	conda-forge
libcamd-3.3.3	h32481e8_7100102	46 KB	conda-forge
libcbblas-3.9.0	31_he106b2a_openblas	16 KB	conda-forge
libccolamd-3.3.4	h32481e8_7100102	42 KB	conda-forge
libcholmod-5.3.1	h59ddab4_7100102	1.1 MB	conda-forge
libcolamd-3.3.4	h32481e8_7100102	33 KB	conda-forge
libcxsparse-4.4.1	h32481e8_7100102	118 KB	conda-forge
libgcc-14.2.0	h767d61c_2	828 KB	conda-forge
libgcc-ng-14.2.0	h69a702a_2	52 KB	conda-forge
libgfortran-14.2.0	h69a702a_2	52 KB	conda-forge
libgfortran-ng-14.2.0	h69a702a_2	53 KB	conda-forge
libgfortran5-14.2.0	hf1ad2bd_2	1.4 MB	conda-forge
libgomp-14.2.0	h767d61c_2	449 KB	conda-forge
libklu-2.3.5	hf24d653_7100102	142 KB	conda-forge
liblapack-3.9.0	31_h7ac8fdf_openblas	16 KB	conda-forge
libldl-3.3.2	h32481e8_7100102	24 KB	conda-forge
libopenblas-0.3.29	pthreads_h94d23a6_0	5.6 MB	conda-forge
libparu-1.0.0	h17147ab_7100102	91 KB	conda-forge
librbio-4.3.4	h32481e8_7100102	47 KB	conda-forge
libspex-3.2.3	had10066_7100102	79 KB	conda-forge
libspqr-4.3.4	h852d39f_7100102	213 KB	conda-forge
libstdcxx-14.2.0	h8f9b012_2	3.7 MB	conda-forge
libstdcxx-ng-14.2.0	h4852527_2	53 KB	conda-forge
libsuitesparseconfig-7.10.1	h92d6892_7100102	42 KB	conda-forge
libumfpack-6.3.5	heb53515_7100102	424 KB	conda-forge
metis-5.1.0	hd0bcaf9_1007	3.7 MB	conda-forge
mpfr-4.2.1	h90cbb55_3	620 KB	conda-forge

numpy-2.2.4	py311h5d046bc_0	8.6 MB	conda-forge
openssl-3.4.1	h7b32b05_0	2.8 MB	conda-forge
pyfmi-2.16.3	py311h9f3472d_0	5.2 MB	conda-forge
python_abi-3.11	2_cp311	5 KB	conda-forge
scipy-1.15.2	py311h8f841c2_0	16.4 MB	conda-forge
suitesparse-7.10.1	ha0f6916_7100102	12 KB	conda-forge
sundials-7.1.1	ha52427a_0	907 KB	conda-forge

Total:		56.1 MB	

The following NEW packages will be INSTALLED:

_x86_64-microarch~	conda-forge/noarch::_x86_64-microarch-level-3-2_broadwell
assimulo	conda-forge/linux-64::assimulo-3.6.0-py311h083bc19_0
fmilib	conda-forge/linux-64::fmilib-2.4.1-hac33072_1
gmp	conda-forge/linux-64::gmp-6.3.0-hac33072_2
libamd	conda-forge/linux-64::libamd-3.3.3-haaf9dc3_7100102
libblas	conda-forge/linux-64::libblas-3.9.0-31_h59b9bed_openblas
libbtf	conda-forge/linux-64::libbtf-2.3.2-h32481e8_7100102
libcamd	conda-forge/linux-64::libcamd-3.3.3-h32481e8_7100102
libcbblas	conda-forge/linux-64::libcbblas-3.9.0-31_he106b2a_openblas
libccolamd	conda-forge/linux-64::libccolamd-3.3.4-h32481e8_7100102
libcholmod	conda-forge/linux-64::libcholmod-5.3.1-h59ddab4_7100102
libcolamd	conda-forge/linux-64::libcolamd-3.3.4-h32481e8_7100102
libcxsparse	conda-forge/linux-64::libcxsparse-4.4.1-h32481e8_7100102
libgcc	conda-forge/linux-64::libgcc-14.2.0-h767d61c_2
libgfortran	conda-forge/linux-64::libgfortran-14.2.0-h69a702a_2
libgfortran-ng	conda-forge/linux-64::libgfortran-ng-14.2.0-h69a702a_2
libgfortran5	conda-forge/linux-64::libgfortran5-14.2.0-hf1ad2bd_2
libklu	conda-forge/linux-64::libklu-2.3.5-hf24d653_7100102
liblapack	conda-forge/linux-64::liblapack-3.9.0-31_h7ac8fdf_openblas
libldl	conda-forge/linux-64::libldl-3.3.2-h32481e8_7100102
libopenblas	conda-forge/linux-64::libopenblas-0.3.29-pthreads_h94d23a6_0
libparu	conda-forge/linux-64::libparu-1.0.0-h17147ab_7100102
librbio	conda-forge/linux-64::librbio-4.3.4-h32481e8_7100102
libspex	conda-forge/linux-64::libspex-3.2.3-had10066_7100102
libspqr	conda-forge/linux-64::libspqr-4.3.4-h852d39f_7100102
libstdcxx	conda-forge/linux-64::libstdcxx-14.2.0-h8f9b012_2
libsuitesparsecon~	conda-forge/linux-64::libsuitesparseconfig-7.10.1-h92d6892_7100102
libumfpack	conda-forge/linux-64::libumfpack-6.3.5-heb53515_7100102
metis	conda-forge/linux-64::metis-5.1.0-hd0bc9f9_1007
mpfr	conda-forge/linux-64::mpfr-4.2.1-h90cbb55_3
numpy	conda-forge/linux-64::numpy-2.2.4-py311h5d046bc_0
pyfmi	conda-forge/linux-64::pyfmi-2.16.3-py311h9f3472d_0
python_abi	conda-forge/linux-64::python_abi-3.11-2_cp311
scipy	conda-forge/linux-64::scipy-1.15.2-py311h8f841c2_0
suitesparse	conda-forge/linux-64::suitesparse-7.10.1-ha0f6916_7100102
sundials	conda-forge/linux-64::sundials-7.1.1-ha52427a_0

The following packages will be UPDATED:

conda	pkgs/main::conda-24.11.1-py311h06a430~ --> conda-forge::conda-25.3.0-py311h38be061_0
libgcc-ng	pkgs/main::libgcc-ng-11.2.0-h1234567_1 --> conda-forge::libgcc-ng-14.2.0-h69a702a_2

```
libgomp                pkgs/main::libgomp-11.2.0-h1234567_1 --> conda-forge::libgomp
-14.2.0-h767d61c_2
libstdcxx-ng           pkgs/main::libstdcxx-ng-11.2.0-h12345~ --> conda-forge::libstdc
xx-ng-14.2.0-h4852527_2
openssl                pkgs/main::openssl-3.0.16-h5eee18b_0 --> conda-forge::openssl
-3.4.1-h7b32b05_0
```

The following packages will be SUPERSEDED by a higher-priority channel:

```
certifi                pkgs/main/linux-64::certifi-2025.1.31~ --> conda-forge/noarch::
certifi-2025.1.31-pyhd8ed1ab_0
```

Downloading and Extracting Packages:

```
scipy-1.15.2           | 16.4 MB | : 0% 0/1 [00:00<?, ?it/s]
numpy-2.2.4            | 8.6 MB  | : 0% 0/1 [00:00<?, ?it/s]
```

```
libopenblas-0.3.29    | 5.6 MB  | : 0% 0/1 [00:00<?, ?it/s]
```

```
pyfmi-2.16.3          | 5.2 MB  | : 0% 0/1 [00:00<?, ?it/s]
```

```
metis-5.1.0           | 3.7 MB  | : 0% 0/1 [00:00<?, ?it/s]
```

```
libstdcxx-14.2.0      | 3.7 MB  | : 0% 0/1 [00:00<?, ?it/s]
```

```
openssl-3.4.1         | 2.8 MB  | : 0% 0/1 [00:00<?, ?it/s]
```

```
libgfortran5-14.2.0   | 1.4 MB  | : 0% 0/1 [00:00<?, ?it/s]
```

```
conda-25.3.0          | 1.1 MB  | : 0% 0/1 [00:00<?, ?it/s]
```

assimulo-3.6.0 | 1.1 MB | : 0% 0/1 [00:00<?, ?it/s]

libcholmod-5.3.1 | 1.1 MB | : 0% 0/1 [00:00<?, ?it/s]

sundials-7.1.1 | 907 KB | : 0% 0/1 [00:00<?, ?it/s]

libgcc-14.2.0 | 828 KB | : 0% 0/1 [00:00<?, ?it/s]

mpfr-4.2.1 | 620 KB | : 0% 0/1 [00:00<?, ?it/s]

gmp-6.3.0 | 449 KB | : 0% 0/1 [00:00<?, ?it/s]

libgomp-14.2.0 | 449 KB | : 0% 0/1 [00:00<?, ?it/s]

libumfpack-6.3.5 | 424 KB | : 0% 0/1 [00:00<?, ?it/s]

fmilib-2.4.1	383 KB	:	0% 0/1 [00:00<?, ?it/s]
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libspqr-4.3.4	213 KB	:	0% 0/1 [00:00<?, ?it/s]
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... (more hidden) ...			
scipy-1.15.2	16.4 MB	:	0% 0.0009529389827073913/1 [00:00<02:08, 128.60s/it]

pyfmi-2.16.3	5.2 MB	:	0% 0.002983953056648666/1 [00:00<00:39, 39.68s/it]
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metis-5.1.0	3.7 MB	:	0% 0.004175799528999174/1 [00:00<00:29, 29.35s/it]
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libopenblas-0.3.29	5.6 MB	:	0% 0.0027679004637044184/1 [00:00<00:48, 48.19s/it]
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scipy-1.15.2	16.4 MB	:	9% 0.09052920335720217/1 [00:00<00:01, 2.07s/it]
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pyfmi-2.16.3 t/s]	5.2 MB	: 26% 0.2566199628717853/1 [00:00<00:00, 1.39i
metis-5.1.0 t/s]	3.7 MB	: 47% 0.4676895472479075/1 [00:00<00:00, 2.51i
libopenblas-0.3.29 t/s]	5.6 MB	: 29% 0.2878616482252595/1 [00:00<00:00, 1.39i
scipy-1.15.2 t/s]	16.4 MB	: 32% 0.3182816202242687/1 [00:00<00:00, 1.26i
libopenblas-0.3.29 t/s]	5.6 MB	: 96% 0.9576935604417288/1 [00:00<00:00, 3.56i
metis-5.1.0	3.7 MB	: 100% 1.0/1 [00:00<00:00, 3.31it/s]
scipy-1.15.2 t/s]	16.4 MB	: 55% 0.5507987320048722/1 [00:00<00:00, 1.66i
libstdcxx-14.2.0 5s/it]	3.7 MB	: 0% 0.0042177278432850495/1 [00:00<01:38, 98.4
pyfmi-2.16.3	5.2 MB	: 100% 1.0/1 [00:00<00:00, 2.79it/s]
pyfmi-2.16.3	5.2 MB	: 100% 1.0/1 [00:00<00:00, 2.79it/s]
openssl-3.4.1 6s/it]	2.8 MB	: 1% 0.0055741049077571376/1 [00:00<01:27, 88.1
libopenblas-0.3.29	5.6 MB	: 100% 1.0/1 [00:00<00:00, 3.56it/s]
scipy-1.15.2 t/s]	16.4 MB	: 72% 0.7223277488922026/1 [00:00<00:00, 1.61i

libgfortran5-14.2.0	1.4 MB	: 1% 0.011206734985068174/1 [00:00<00:50, 50.66s/it]
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scipy-1.15.2	16.4 MB	: 92% 0.9195861183126326/1 [00:00<00:00, 1.71it/s]
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libgfortran5-14.2.0	1.4 MB	: 100% 1.0/1 [00:00<00:00, 50.66s/it]
numpy-2.2.4	8.6 MB	: 100% 1.0/1 [00:00<00:00, 1.47it/s]
numpy-2.2.4	8.6 MB	: 100% 1.0/1 [00:00<00:00, 1.47it/s]

openssl-3.4.1	2.8 MB	: 100% 1.0/1 [00:00<00:00, 2.23it/s]
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libstdc++-14.2.0	3.7 MB	: 100% 1.0/1 [00:00<00:00, 1.72it/s]
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conda-25.3.0	1.1 MB	: 1% 0.01395284272989957/1 [00:00<00:49, 50.65s/it]
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libcholmod-5.3.1	1.1 MB	: 1% 0.014870549794649543/1 [00:00<00:48, 49.66s/it]
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conda-25.3.0 | 1.1 MB | : 100% 1.0/1 [00:00<00:00, 50.65s/it]

metis-5.1.0 | 3.7 MB | : 100% 1.0/1 [00:00<00:00, 3.31it/s]

libcholmod-5.3.1 | 1.1 MB | : 100% 1.0/1 [00:00<00:00, 49.66s/it]

sundials-7.1.1 | 907 KB | : 2% 0.01763373830085844/1 [00:00<00:44, 45.65
s/it]

libgcc-14.2.0 | 828 KB | : 2% 0.01932337522187561/1 [00:00<00:42, 43.11
s/it]

mpfr-4.2.1 | 620 KB | : 3% 0.025811696239942908/1 [00:00<00:31, 32.63
s/it]

pyfmi-2.16.3 | 5.2 MB | : 100% 1.0/1 [00:00<00:00, 2.79it/s]

scipy-1.15.2 | 16.4 MB | : 100% 1.0/1 [00:00<00:00, 1.71it/s]

assimulo-3.6.0 | 1.1 MB | : 1% 0.014703493605362324/1 [00:00<00:58, 59.20
s/it]

mpfr-4.2.1 | 620 KB | : 100% 1.0/1 [00:00<00:00, 32.63s/it]

libgcc-14.2.0 | 828 KB | : 100% 1.0/1 [00:00<00:00, 43.11s/it]

gmp-6.3.0 | 449 KB | : 4% 0.03561313321233331/1 [00:00<00:24, 25.83
s/it]

libumfpack-6.3.5 | 424 KB | : 4% 0.037731330084655984/1 [00:00<00:23, 24.74
s/it]

gmp-6.3.0 | 449 KB | : 100% 1.0/1 [00:00<00:00, 25.83s/it]

libgomp-14.2.0 | 449 KB | : 4% 0.03562807972826631/1 [00:00<00:25, 26.40
s/it]

libumfpack-6.3.5 | 424 KB | : 100% 1.0/1 [00:00<00:00, 24.74s/it]

libgomp-14.2.0 | 449 KB | : 100% 1.0/1 [00:00<00:00, 26.40s/it]

assimulo-3.6.0 | 1.1 MB | : 21% 0.20584891047507253/1 [00:00<00:02, 3.53
s/it]

fmilib-2.4.1 | 383 KB | : 4% 0.04180391656566945/1 [00:00<00:22, 23.66

s/it]

libspqr-4.3.4 | 213 KB | : 8% 0.07503068271326775/1 [00:00<00:12, 13.26
s/it]

fmilib-2.4.1 | 383 KB | : 100% 1.0/1 [00:01<00:00, 23.66s/it]

... (more hidden) ...

libgfortran5-14.2.0 | 1.4 MB | : 100% 1.0/1 [00:01<00:00, 1.17it/s]

libgfortran5-14.2.0 | 1.4 MB | : 100% 1.0/1 [00:01<00:00, 1.17it/s]

... (more hidden) ...

libspqr-4.3.4 | 213 KB | : 100% 1.0/1 [00:01<00:00, 13.26s/it]

assimulo-3.6.0	1.1 MB	: 76% 0.7645816674788408/1 [00:01<00:00, 1.15it/s]
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assimulo-3.6.0	1.1 MB	: 100% 1.0/1 [00:01<00:00, 1.15it/s]
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libopenblas-0.3.29	5.6 MB	: 100% 1.0/1 [00:01<00:00, 3.56it/s]
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openssl-3.4.1	2.8 MB	: 100% 1.0/1 [00:01<00:00, 2.23it/s]
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libstdcxx-14.2.0	3.7 MB	: 100% 1.0/1 [00:01<00:00, 1.72it/s]
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conda-25.3.0	1.1 MB	: 100% 1.0/1 [00:02<00:00, 2.03s/it]
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conda-25.3.0	1.1 MB	: 100% 1.0/1 [00:02<00:00, 2.03s/it]
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libcholmod-5.3.1	1.1 MB	: 100% 1.0/1 [00:02<00:00, 2.06s/it]
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libcholmod-5.3.1	1.1 MB	: 100% 1.0/1 [00:02<00:00, 2.06s/it]
------------------	--------	--------------------------------------

sundials-7.1.1	907 KB	: 100% 1.0/1 [00:02<00:00, 2.21s/it]
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sundials-7.1.1	907 KB	: 100% 1.0/1 [00:02<00:00, 2.21s/it]
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mpfr-4.2.1	620 KB	: 100% 1.0/1 [00:02<00:00, 2.27s/it]
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numpy-2.2.4	8.6 MB	: 100% 1.0/1 [00:02<00:00, 1.47it/s]

libgcc-14.2.0 | 828 KB | : 100% 1.0/1 [00:02<00:00, 2.49s/it]

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gmp-6.3.0 | 449 KB | : 100% 1.0/1 [00:02<00:00, 2.50s/it]

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libumfpack-6.3.5 | 424 KB | : 100% 1.0/1 [00:02<00:00, 2.51s/it]

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libgomp-14.2.0 | 449 KB | : 100% 1.0/1 [00:02<00:00, 2.58s/it]

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fmilib-2.4.1	383 KB	: 100% 1.0/1 [00:02<00:00, 2.64s/it]
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libspqr-4.3.4	213 KB	: 100% 1.0/1 [00:02<00:00, 2.70s/it]
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libspqr-4.3.4	213 KB	: 100% 1.0/1 [00:02<00:00, 2.70s/it]
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scipy-1.15.2 | 16.4 MB | : 100% 1.0/1 [00:04<00:00, 1.71it/s]


```
Preparing transaction: - 00\ 00done
Verifying transaction: / 00- 00\ 00| 00/ 00done
Executing transaction: \ 00| 00/ 00- 00\ 00| 00/ 00- 00\ 00| 00/ 00- 00\ 00| 00/ 00- 00\ 00| 00/ 00- 00\ 00| 00/ 00- 00\ 00done
```

BPL STEM_AIR_Perfusion setup

Now specific installation and the run simulations. Start with connecting to Github. Then upload the two files:

- FMU - BPL_STEM_AIR_Perfusion_linux_om_me.fmu
- Setup-file - BPL_STEM_AIR_Perfusion_explore

```
In [9]: # Filter out DeprecationWarnings for 'np.float as alias' is needed - wish I could m
import warnings
```

```
warnings.filterwarnings("ignore")
```

```
In [10]: %%bash
git clone https://github.com/janpeter19/BPL_STEM_AIR_Perfusion
```

```
Cloning into 'BPL_STEM_AIR_Perfusion'...
```

```
In [11]: %cd BPL_STEM_AIR_Perfusion
/content/BPL_STEM_AIR_Perfusion
```

BPL_STEM_AIR_Perfusion - test

Author: Jan Peter Axelsson

Here we show simulations of stem cell cultivation in an aerated hollow fiber reactor. The reactor volume is kept constant and cells recycled, thus the setup is similar to perfusion cultivation.

The model combines rudimentary cell growth and metabolism combined with times series data of the metabolic rates: q_{Nmax} , q_{Lc} , and q_{O2} , marked with red in the comprehensive plot.

Ref Greuel et al: "Online measurement of oxygen enables continuous noninvasive evaluation of human-induced pluripotent stem cell (hiPSC) culture in a perfused 3D hollow-fiber bioreactor", Biotech. Bioeng., 2019.

Setup of the simulation model

```
In [12]: run -i BPL_STEM_AIR_Perfusion_explore.py
```

Linux - run FMU pre-compiled OpenModelica

Model for the process has been setup. Key commands:

- par() - change of parameters and initial values
- init() - change initial values only
- simu() - simulate and plot
- newplot() - make a new plot
- show() - show plot from previous simulation
- disp() - display parameters and initial values from the last simulation
- describe() - describe culture, broth, parameters, variables with values/units

Note that both `disp()` and `describe()` takes values from the last simulation and the command `process_diagram()` brings up the main configuration

Brief information about a command by `help()`, eg `help(simu)`

Key system information is listed with the command `system_info()`

```
In [13]: %matplotlib inline
plt.rcParams['figure.figsize'] = [30/2.54, 24/2.54]
```

About the process model

Here a process diagram is shown of the process. Further information about the culutre stored in the model code is extracted.

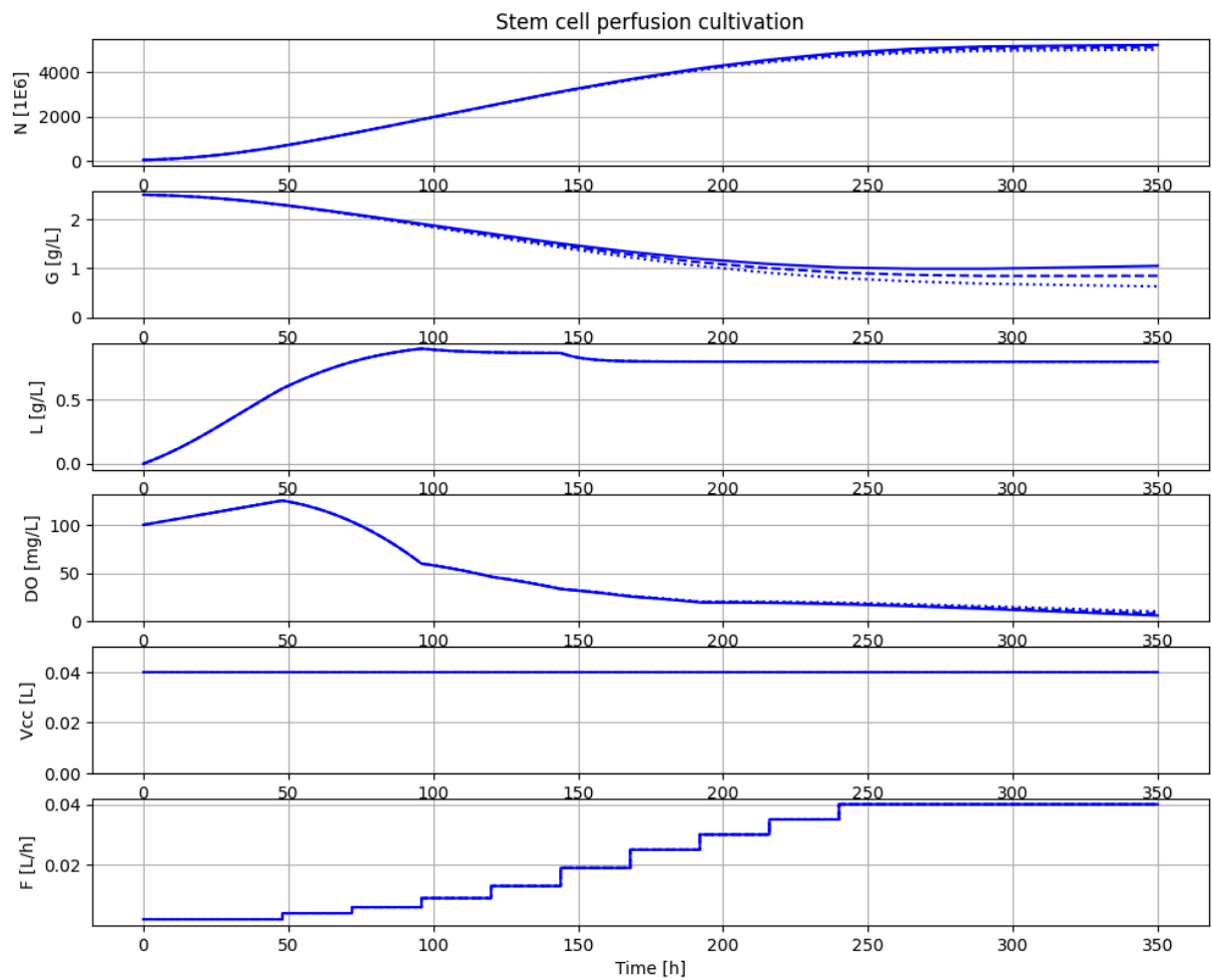
```
In [14]: #process_diagram()
```

Simulations

```
In [15]: # Process parameters to mimic stem cell example  
# - study impact of different values maintenatnce qm
```

```
newplot(plotType='Basic')  
for value in [1.0e-6, 10.0e-6, 20e-6]:  
    par(qm=value)  
    simu(350)
```

```
Could not find cannot import name 'dopri5' from 'assimulo.lib' (/usr/local/lib/pytho  
n3.11/site-packages/assimulo/lib/__init__.py)  
Could not find cannot import name 'rodas' from 'assimulo.lib' (/usr/local/lib/python  
3.11/site-packages/assimulo/lib/__init__.py)  
Could not find cannot import name 'odassl' from 'assimulo.lib' (/usr/local/lib/pytho  
n3.11/site-packages/assimulo/lib/__init__.py)  
Could not find ODEPACK functions.  
Could not find RADAR5  
Could not find GLIMDA.
```

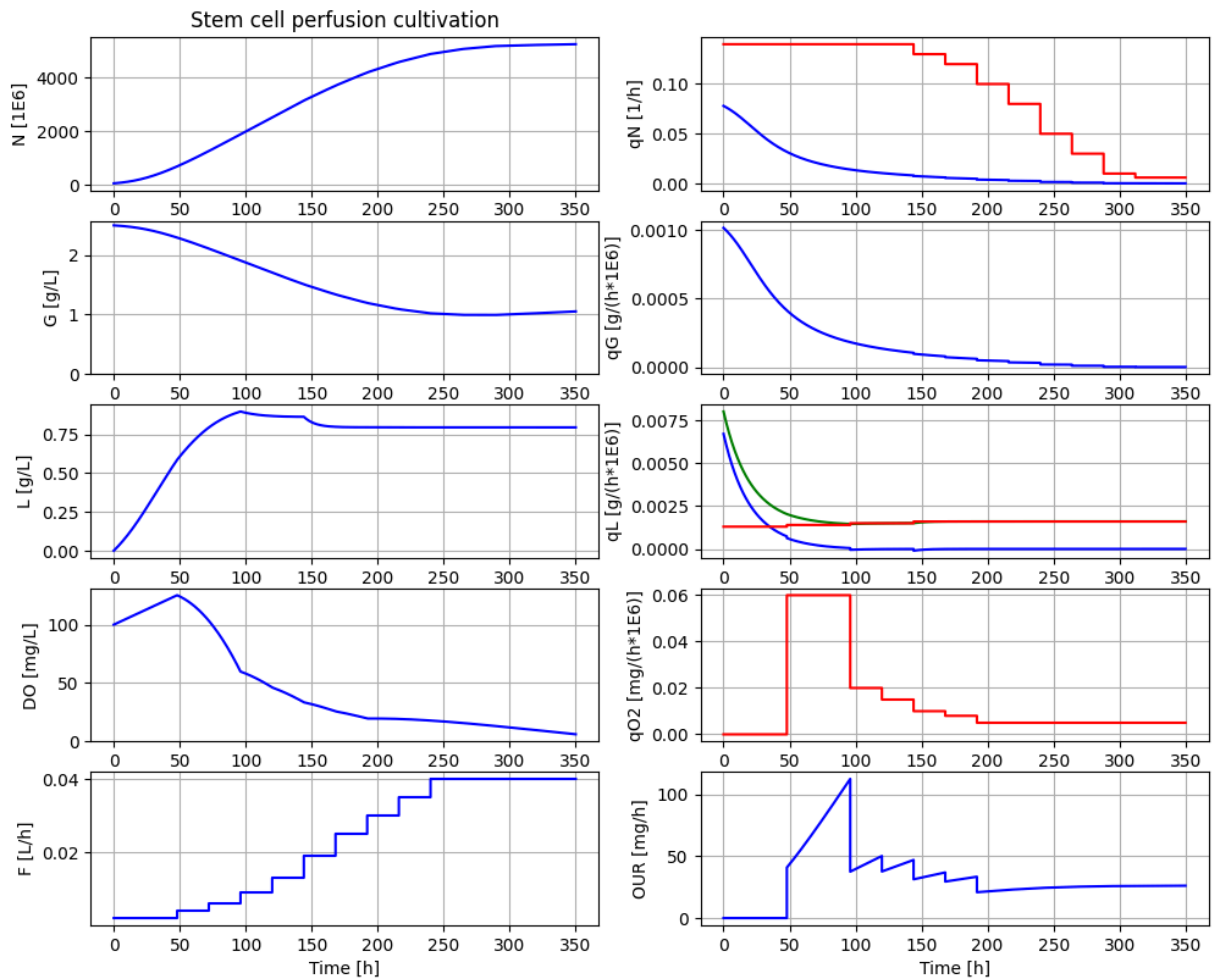



In [16]: *# Process parameters to mimic stem cell example*
. comprehensvie plot including metabolic rates

```
par(qm=1e-6)
```

```
newplot(plotType='Comprehensive')
```

```
simu(350)
```



In [17]: `describe('culture')`

Reactor culture human-induced pluripotent stem cells - hiPSCs

In [18]: `describe('Vcc')`

Volume of the reactor : 0.04 [L]

In [19]: `describe('N_start')`

Number of cells at start : 50.0 [1E6]

In [20]: `describe('N')`

Number of cells : 5223.752 [1E6]

In [21]: `describe('G_start')`

Glucose conc at start : 2.5 [g/L]

In [22]: `describe('G_in')`

Glucose feed conc : 2.5 [g/L]

In [23]: `describe('L')`

Lactate conc : 0.796 [g/L]

In [24]: `describe('DO')`

Dissolved oxygen conc : 6.044 [mg/L]

Appendix

In [25]: `system_info()`

System information

-OS: Linux

-Python: 3.11.11

-Scipy: not installed in the notebook

-PyFMI: 2.16.3

-FMU by: OpenModelica Compiler OpenModelica 1.25.0~dev-51-ge672d09

-FMI: 2.0

-Type: FMUModelME2

-Name: BPL_STEM.Reactor

-Generated: 2024-10-03T11:45:51Z

-MSL: 3.2.3

-Description: BPL - not used

-Interaction: FMU-explore version 1.0.0

In [25]: